



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Spring 2023

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1.	Read the following code and detect different types of errors. <pre>1. #include <studio.h> 2. double commite (floar april){ 3. float b=5; 4. int b, c; 5. b = summation(int c,,float aprjl) { 6. retum 1; 7. } 8. void summation(int b){ 9. Float k == b*2; 10. retun b; 11. } 12. int main(){ 13. int b=2,a=2, 14. float c; 15. if(B==2) 16. c = commitee(50); 17. a = c+2; 18. return 0; 19. }</pre>	8
2.	(a) Generate Predictive Parsing Table for the following grammar with FIRST and FOLLOW sets. Consider all the terminals are of single character/number/symbol. <pre>X → YZ abc bZ Z → +YWa abYa b Y → RWa bY c ∈ W → Rc baZW b a R → c [X] * WZ ∈</pre>	[4+4]

	(b) Construct a Predictive Parsing table for the following grammar & also check whether string $id + (id * id)$ is accepted or not. The sequence of Moves by Predictive Parser in a table (should have column: stack,input, action). $E \rightarrow E + E \mid (E) \mid A$ $A \rightarrow A * A \mid AB \mid \epsilon$ $B \rightarrow id \mid \epsilon$	
3.	For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. Consider all the terminals are of single character/number/symbol.	[4+4]
	(a) $S \rightarrow AuStrALia \mid dHaKA$ $A \rightarrow CanAdA \mid LoKDoH \mid \epsilon$ $L \rightarrow SDiLanKa \mid iKaC \mid \epsilon$ $H \rightarrow AfriCa \mid uSA \mid \epsilon$ $K \rightarrow LfHAiu \mid xASiu$ $D \rightarrow KiutLAI \mid pLKig$ $C \rightarrow LfHAiu \mid xASiu$ (b) $S \rightarrow ACB \mid CbB \mid Ba$ $A \rightarrow da \mid BC$ $B \rightarrow BB \mid g \mid \epsilon$ $C \rightarrow DB \mid h \mid \epsilon$ $D \rightarrow 1 \mid 2 \mid 3 \mid \epsilon$	
4.	Eliminate immediate left recursion from the following grammar.	[4+4]
	(a) $X \rightarrow xa \mid +Xai \mid YBC \mid X + 2 \mid [x - 2] \mid Xikl$ $Y \rightarrow YYaB \mid iEhb \mid Yap(g) \mid abc \mid YaE \mid CE$ $A \rightarrow 0A(0) \mid AB02 \mid 2 * 3A \mid A[i * 2] \mid (A + 2)$ $B \rightarrow B \mid E ig \mid Dxy \mid BCy \mid ByC \mid xyB \mid BC$ (b) $I \rightarrow I + Z \mid I + X \mid I * Y + Z \mid /IY$ $J \rightarrow Jli \mid Kl * 2 \mid j/Z \mid J + 2[3]$ $Z \rightarrow (Z) \mid Z + (X * Y) \mid Z + 23$ $F \rightarrow 23 \mid Ftu \mid Y[F] \mid F(XYZ)$	
5.	Left Factor the following. (a) Consider each word as a single terminal. $P \rightarrow Birds \text{ are beautiful} \mid Birds \text{ Sings the Song} \mid Birds \text{ Singing sounds nice} \mid Flowers \mid flowers \text{ looks nice} \mid flowers \text{ lovely} \mid flowers \text{ lovers} \mid Birds \text{ are small} \mid Birpt \mid Bird \text{ flies}$	[4+4]
	(b) Consider all the terminals are of single character/number/symbol. $A \rightarrow 123456 \mid 121415 \mid 1134214 \mid 112256 \mid 323456 \mid 321156 \mid 32155 \mid 3234156$ $B \rightarrow g+h \mid g+(y) \mid g - ((x+y))+2 \mid (p+q)*(a-b) \mid (u-v)-2 \mid g-x \mid (u+y) \mid (p+x) \mid (p+q)/24 \mid (u + gh)$	